

Development, Bug Fixes, and Release Quality

Code navigation

The backend entry point and route registry are `backend/src/routes/index.ts` and `backend/src/index.ts`. The common backend flow is:

```
route → middleware/validation/authz → controller → service → Prisma/database
```

The frontend route registry is `frontend/App.tsx`. The usual frontend flow is:

```
page/component → domain service in frontend/src/services/ → Axios API client → backend endpoint
```

Shared UI and context live under `frontend/src/components/` and `frontend/src/context/`. Main operational pages are under `frontend/pages/`; authentication pages are under `frontend/src/pages/`.

Safe change workflow

- 1 Reproduce the issue with a focused test, API call, or browser scenario.
- 2 Trace the complete contract: UI → service → route → controller → service → model/migration.
- 3 Identify authorization, tenant scope, workflow state, audit, notification, and concurrency effects before editing.
- 4 Add or update a regression test at the narrowest meaningful layer.
- 5 Implement the smallest change in the existing layer; keep controllers thin.
- 6 Run focused tests, then the package build and relevant broader suite.
- 7 Perform a browser smoke test for user-visible changes.
- 8 Review `git diff`, `git diff --check`, and changed files for secrets or unrelated edits.
- 9 Record migration, seed, feature-flag, and rollout implications in the change/PR description.

Common fix patterns

Request/workflow status bug

Start with `backend/src/services/requestTransition.service.ts`, `availableTransitions.service.ts`, workflow guards, and the published workflow records. Do not scatter direct `prisma.request.update({ status })` calls. The workflow designer/database configuration is authoritative at runtime; the hardcoded fallback map is not the normal control surface.

Permission or missing-menu bug

Trace the backend `requirePermission()` middleware and permission loading/caching first, then the frontend route/page guard. A visible button is not authorization. Verify both allowed and denied paths, including direct API access.

Wrong or missing records

Check tenant scoping, resource authorization, `deletedAt: null`, entity/department scope, and optimistic version handling. A 404 may be intentional to avoid disclosing an inaccessible record.

Notification bug

Trace notification creation, durable outbox dispatch, email provider configuration, SSE/Redis pub-sub, and frontend subscription/reconnect behavior. Verify both the durable in-app record and delivery path.

Attachment or download bug

Trace upload validation, S3-compatible storage, ClamAV scanning, file status, presigned download route, and access control. Do not enable local upload serving in production as a shortcut.

PDF/export bug

Trace the PDF job route, BullMQ/Redis result key, worker, Puppeteer runtime, and frontend polling. Verify an in-container PDF smoke result rather than only a successful enqueue response.

CRM import bug

Use the CRM-specific validators, controller/service, and import tests. Preserve existing fields and report created/skipped/failed rows. Duplicate activities are rejected using Lead ID + Activity Type + normalized Subject.

Credit bug

Use backend/src/credit/ and its dedicated tests/guides. Respect feature flags, RM/application row scope, MFA, segregation of duties, audit-chain immutability, encrypted fields, and immutable memo/document rules.

Testing matrix

Change	Minimum verification
Backend service/controller	Focused Jest test, `npm run build`, relevant integration test
Prisma schema/migration	Migration test on disposable DB, `prisma migrate status`, backend build, affected integration tests
Frontend page/service	Vitest test, `npm run build`, `npx tsc --noEmit`, browser smoke
Auth/RBAC	Allowed/denied API tests and direct-route browser check
Workflow designer/lifecycle	Published-graph test, transition guard tests, end-to-end request scenario
Notification/SSE/queue	Dispatcher/worker tests and a real local integration smoke
Production deployment change	Full local pre-flight plus deployment runbook gates

Pull request checklist

- ☐ Scope and affected domain are stated.
- ☐ API/frontend contract changes are synchronized.
- ☐ Authorization and tenant/resource scope are tested.
- ☐ Audit, notification, SLA, and state-transition side effects are considered.
- ☐ Migration is reversible or has a documented recovery path.
- ☐ Seed changes are safe and do not prune admin-managed production configuration.
- ☐ Tests/builds are fresh and their commands/results are recorded.
- ☐ No secrets, production dumps, or PII are included.